

Double Pendule

Gaspar Daguet

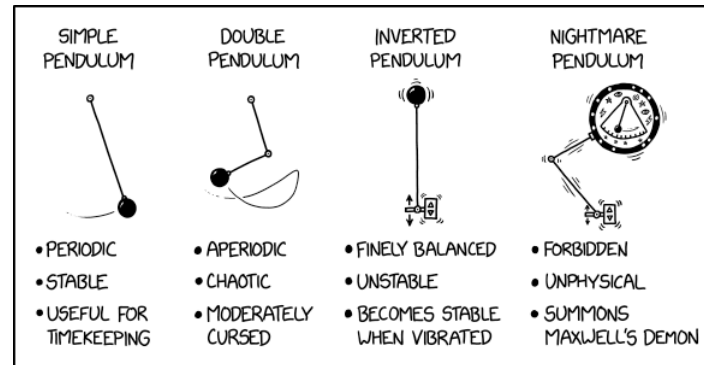


Fig. 1. – Pendulum Types - Xkcd

1. Le TUI
2. Double Pendule
3. Chaîne

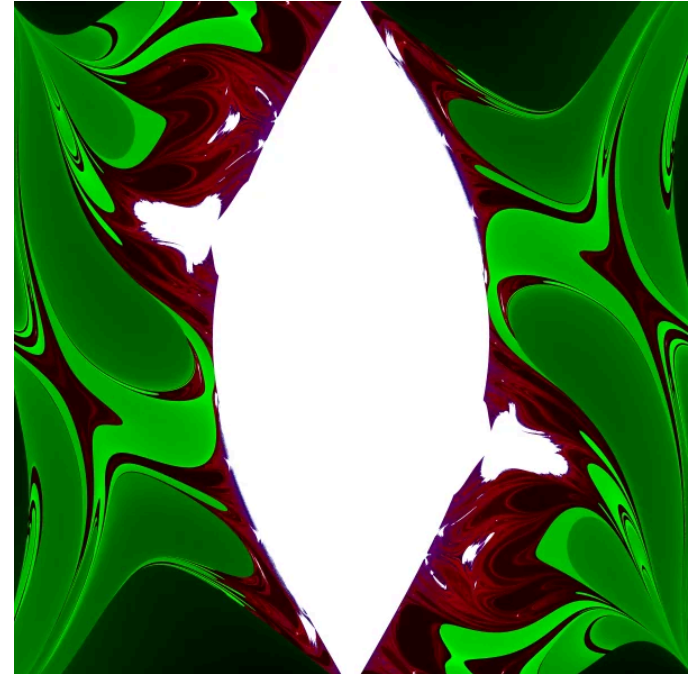
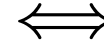
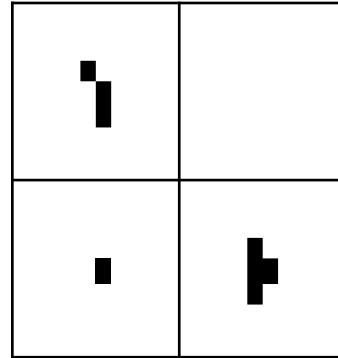
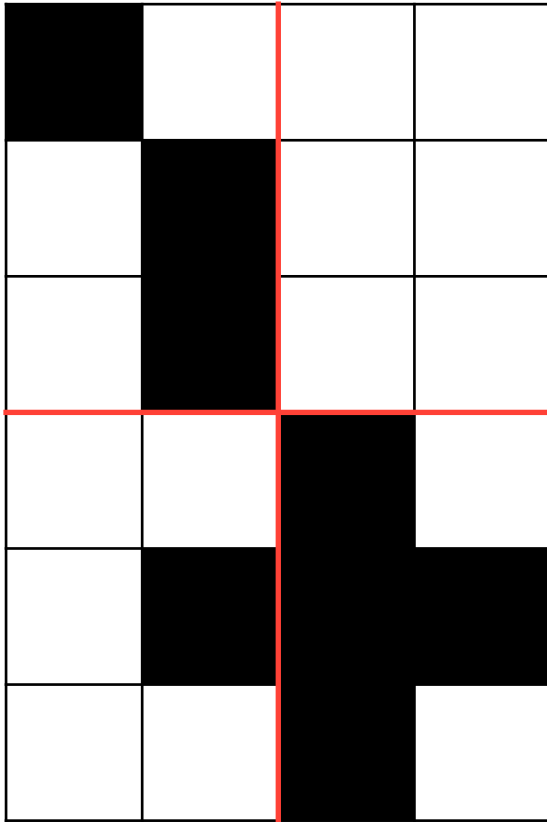


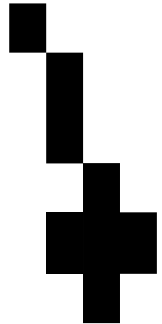
Fig. 2. – Cercle Limite III — M.C. Escher

1) TUI

En machine



ce qui est affiché

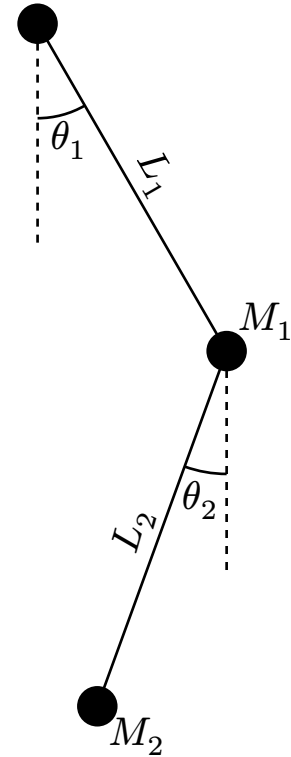


2) Double Pendule

Équation différentielle :

$$(M_1 + M_2)L_1\ddot{\theta}_1 + M_2L_2\ddot{\theta}_2 \cos(\theta_1 - \theta_2) + M_2L_2\dot{\theta}_2^2 \sin(\theta_1 - \theta_2) + (M_1 + M_2)g \sin(\theta_1) = 0$$

$$L_1\ddot{\theta}_1 \cos(\theta_1 - \theta_2) + L_2\ddot{\theta}_2 - M_2L_1\dot{\theta}_1^2 \sin(\theta_1 - \theta_2) + g \sin(\theta_2) = 0$$



$$\Delta\theta_2 = 10^{-6}$$

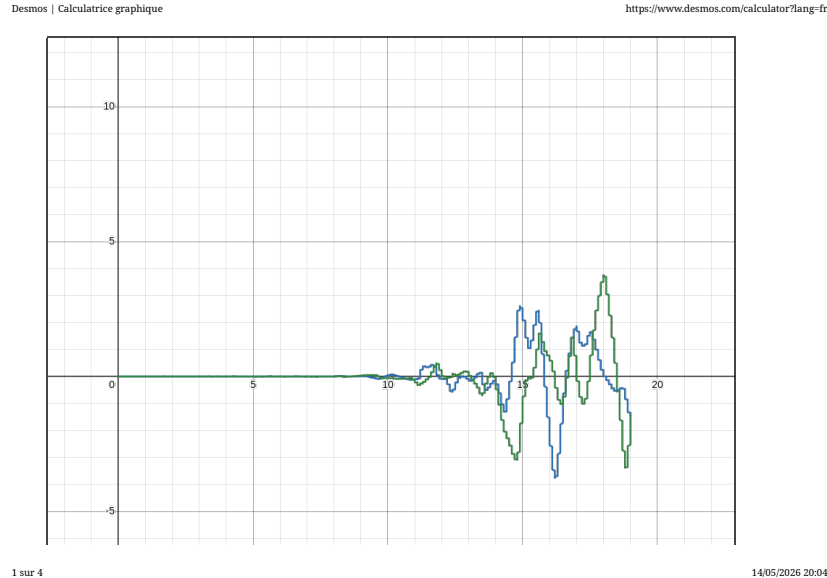


Fig. 3. – Δx et Δy en fonction du temps

pour $\theta_1 = 2.439$
et $\theta_2 = -0.553$

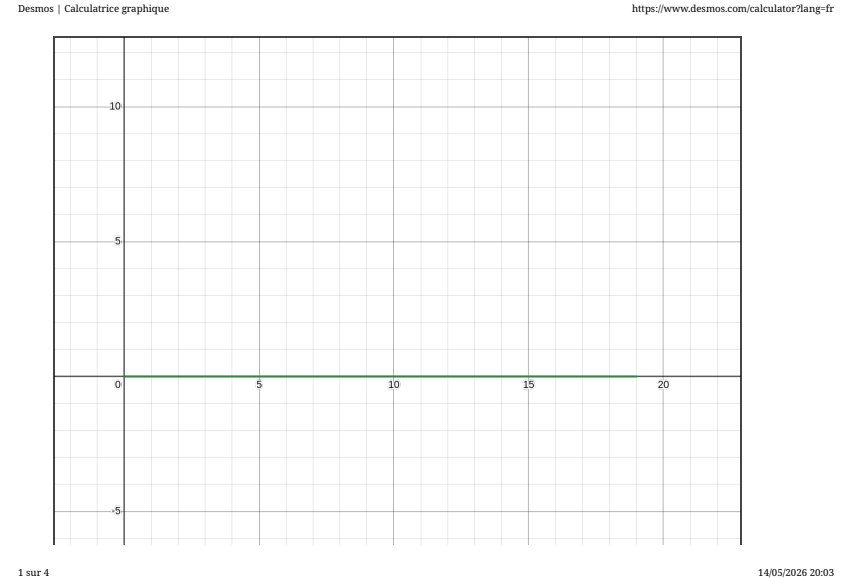
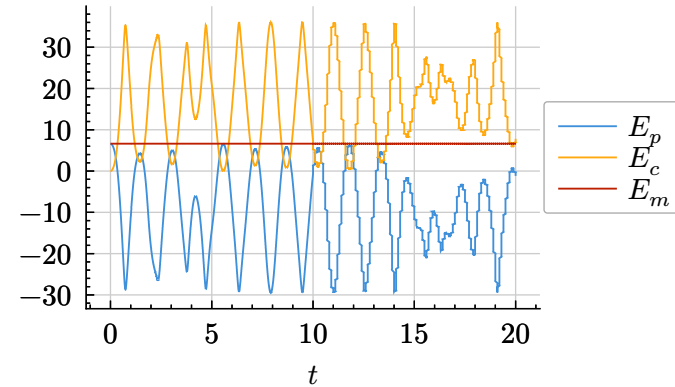
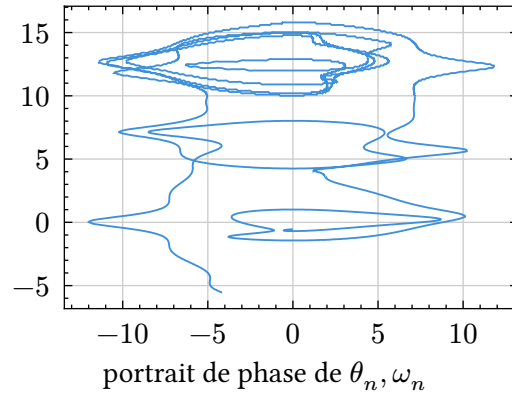


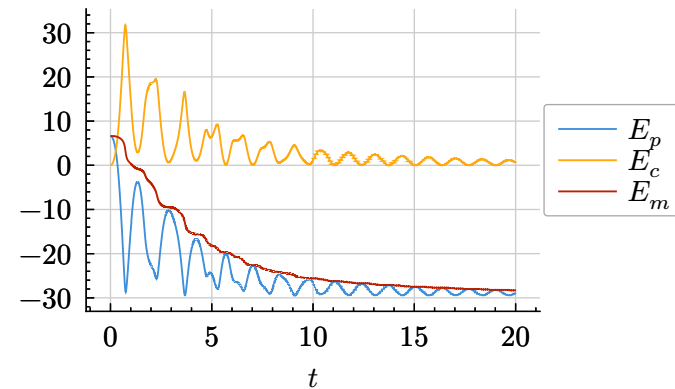
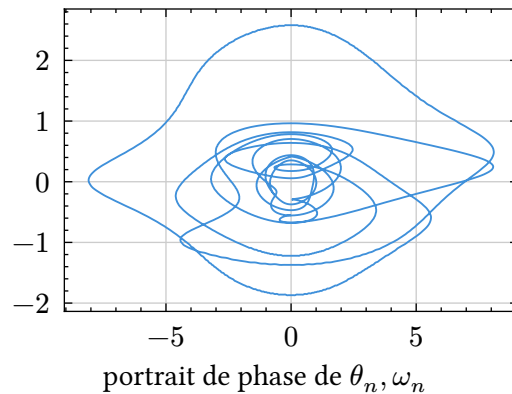
Fig. 4. – Δx et Δy en fonction du temps

pour $\theta_1 = 1.004$
et $\theta_2 = 1.968$

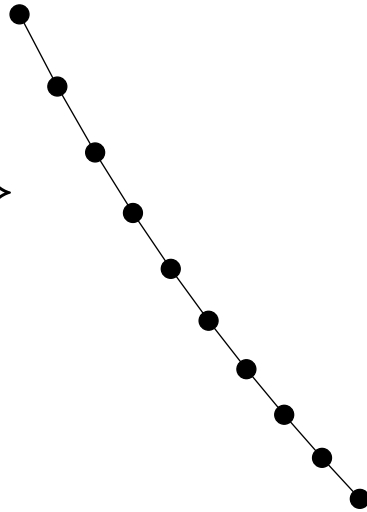
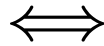
Sans frottement :



Avec frottement :



3) Chaîne



Équation différentielle :

$$\begin{aligned} \forall k \in \llbracket 1; n \rrbracket \\ (n+1-k) \left(\ddot{\theta}_k + \frac{g}{l_k} \sin(\theta_k) \right) \\ + (n+1-k) \sum_{i=1}^{k-1} \left(\ddot{\theta}_i \cos(\theta_k - \theta_i) + \dot{\theta}_i^2 \sin(\theta_k - \theta_i) \right) \\ + \sum_{i=k+1}^n (n+1-j) \left(\ddot{\theta}_i \cos(\theta_i - \theta_k) - \dot{\theta}_i^2 \sin(\theta_i - \theta_k) \right) = 0 \end{aligned}$$

position initiale
de la chaîne :



après quelque
seconde :



position en x
du 50ième et 100ième
pendule :

